PARUL CHAUDHARI

Senior Data Scientist – Fractal AI (9 Years of Experience)

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EDUCATION

- MSc in Machine Learning, Liverpool John Moores University, UK (2021–2022)
- Post Graduate Diploma in ML, IIIT-Bangalore (2020–2021)
- B.Tech in Information Technology, BIET Jhansi (2012–2016)

SKILLS

PROGRAMMING

Python | SQL | PySpark

CLOUD & DATA PLATFORMS

Azure | Databricks | Snowflake | AWS

ML & ANALYTICS

Market Mix Modeling (MMM) | Sales Attribution & Simulation | Budget Optimization | Propensity Modeling | Predictive Modeling | Bayesian Hierarchical Algorithm | Time Series Forecasting | ARIMA | Uplift Modeling | NLP | Deep Learning | GenAI | Agentic AI | LLM Fine-Tuning | Prompt Engineering | Spacy | Causal Inference | XGBoost | LightGBM (LGBM) | Decision Trees | MLOps

TOOLS & VISUALIZATION

Power BI | Excel | PPT | MLflow | Git | SonarQube

SOFT SKILLS

Strategic Storytelling | Stakeholder Engagement | Cross-functional Leadership | Collaborative Problem Solving | Adaptive & Continuous Learning | Data Science Solution Design | Technical Agility (CI/CD)

Career Objective

I aspire to advance human-centric AI by designing intelligent, explainable, and scalable systems that create meaningful real-world impact. With 9 years of experience across data science, analytics, causal ML, conversational AI, and intelligent automation, I am driven by a research-oriented mindset and a deep curiosity for how AI can transform decision-making and productivity. I follow an AEDD philosophy (AI, Engineering, Design, Domain) to build solutions that are not only accurate, but usable, ethical, and outcome-focused, empowering products, people, and businesses for the future.

Fractal AI

Nov 2021 – Present · Bangalore, India

Senior Data Scientist

- Led large-scale **Bayesian MMM, Sales Attribution, and Budget Optimization** initiatives for retail and insurance portfolios, enabling data-driven spend allocation and delivering **~$50M ROI impact** through improved marketing efficiency.
- Delivered **Direct Mail optimization** by combining segmentation, forecasting, and uplift techniques, driving **~$20M savings** by eliminating low-yield volumes and reallocating budgets to high-ROI cohorts.
- Built scalable **Propensity, Forecasting, and MMM pipelines** using Bayesian frameworks, regression, and time-series decomposition to enhance planning accuracy and campaign effectiveness.
- Developed advanced **media impact simulation & planning workflows** (Prophet, Google Meridian) to evaluate channel efficiency, saturation, and lag effects for strategic budget allocation.
- Partnered with **CMOs, marketing leaders, and finance teams** to translate model findings into KPIs and guide quarterly/annual planning.
- Designed **executive-level dashboards and data storytelling**, improving analytics adoption across stakeholders.
- Mentored a team of **4–5 data scientists**, ensuring delivery excellence and capability uplift.
- Drove **direct mail targeting strategies** using customer behavior signals to improve response quality and reduce wasteful frequency.

Data Scientist

- Built and operationalized **insurance-focused MMM components** and **Sales Forecasting** pipelines to quantify channel effectiveness, inform budget allocation, and improve planning accuracy.
- Built and deployed **full-stack ML pipelines on Azure Databricks**, integrated with CI/CD standards (Sonar, GitOps, automated deployments).
- Developed **Product Keyword Ranking** models to improve online visibility and drive organic engagement.

Data Scientist – VMware

July 2019 – Oct 2021 · Pune, India

- Developed **Product Review Sentiment Analysis** (ML/NLP) to classify customer feedback and surface actionable insights for product teams.
- Built **OCR-based PDF Information Extraction** pipelines to improve data accessibility and reduce manual effort.
- Designed and deployed an **Information Retrieval Chatbot** (Rasa + Microsoft tools) for Sales, accelerating access to product information.

ACHIEVEMENTS

- Winner – VMware MLPO'21 Hackathon (Anomaly Detection)
- Top 10 – VMware Product Review Classification MLPO'20
- Spot Awards at Fractal & VMware for complex project delivery
- INSPIRE Scholarship Holder for Innovation in Science
- Mentored IITians on real-world AI/ML use cases
- Core Member – Analytics Vidhya Data Analytics Community

- Applied ML for **anomaly detection, classification, and automation** in internal enterprise products.
- Delivered **predictive analytics for internal product reviews**, improving decision-making and product planning.
- **Winner – VMware MLPO'21 Hackathon (Anomaly Detection)**; **Top 10 – MLPO'20 Classification Challenge**.
- Mentored new hires; improved productivity via automation tools and ML workflows for product engineering teams.

Software Developer – Accenture

July 2017 – June 2019 · Gurugram, India

- Automated repetitive business processes with **RPA frameworks**.
- Delivered early-stage **data science solutions** (Python, Pandas, Matplotlib) to improve internal efficiency.
- Supported consulting projects with emphasis on **process optimization** and stakeholder engagement.